MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. The roots of $(x+1)(x-4)$ are

   A. 1 and 4
   B. -1 and -4
   C. -1 and 4
   D. 1 and -4

2. The discriminant $x^2 - 2x + 3$ is

   A. -8
   B. 8
   C. 16
   D. -2

3. The vertex of $(x-2)^2 + 5$ is

   A. $(2;-5)$
   B. $(-2;5)$
   C. $(5;2)$
   D. $(2;5)$

4. The sum of the roots of 2 $x^2 - 6x + 1$ is

   A. 6
   B. 3
   C. -3
   D. $\frac{1}{2}$

5. $x^2 - 4 < 0$ is equivalent to

   A. $x > 2$
   B. $x < -2$
   C. $x \in]-2;2[$
   D. $x \in [-2;2]$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $x^2 + 6x + 9$ factorises as

   A. $(x-3)^2$
   B. $(x+9)(x+1)$
   C. $(x+6)^2$
   D. $(x+3)^2$

2. The product of the roots of 3 $x^2 + x - 2$ is

   A. $-\frac{2}{3}$

B. $-\frac{1}{3}$

C. $\frac{2}{3}$

D. -2

3. $-x^2 + 4x$ reaches its maximum on [0; 1] at

A. 2

B. 0

C. 1

D. 4

4. A double root requires

A. $a = 0$

B. $\Delta = 0$

C. $b = 0$

D. $c = 0$

5. The quadratic polynomials with roots 0 and 2 are

A. $a(x + 2)$

B. $a(x - 2)^2$

C. $x(x - 2)$ only

D. $ax(x - 2)$, $a \neq 0$

True or false? Justify.

1. A quadratic function always has two real roots.

2. A quadratic with two distinct roots is negative between them.

3. For every real x, ($x - 2)^2 \geq 0$.

Personal review

I justify the choice of form; I distinguish roots and vertex; I finish an inequality with a together; I respect the domain of the model.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.